

SkyCast

Project Documentation

Project Title:

SkyCast – Live Weather Forecast Application

Submitted By:

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Intern ID:

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Organization:

Codtech IT Solutions Pvt. Ltd.

Domain:

Frontend Web Development

Duration:

4 Weeks

Submitted as part of the
Frontend Web Development Internship
at
Codtech IT Solutions Pvt. Ltd.

Project Overview

SkyCast is a responsive weather forecasting web application developed using HTML, CSS, and JavaScript. The application provides real-time weather information for cities around the world through OpenWeatherMap API integration. It features an interactive and modern interface with live weather updates, hourly and weekly forecasts, weather insights, customizable settings, and an interactive world map. The application is designed to deliver an engaging, informative, and user-friendly weather experience across desktop and mobile devices.

Objectives

- Develop a modern weather forecasting web application.
- Display accurate real-time weather information.
- Integrate live weather data using external APIs.
- Provide hourly and weekly weather forecasts.
- Create a responsive and visually appealing interface.
- Improve frontend development and JavaScript skills.

Technologies Used

- HTML5
- CSS3
- JavaScript
- OpenWeatherMap API
- Leaflet.js
- Chart.js
- Local Storage
- Responsive Web Design

Features

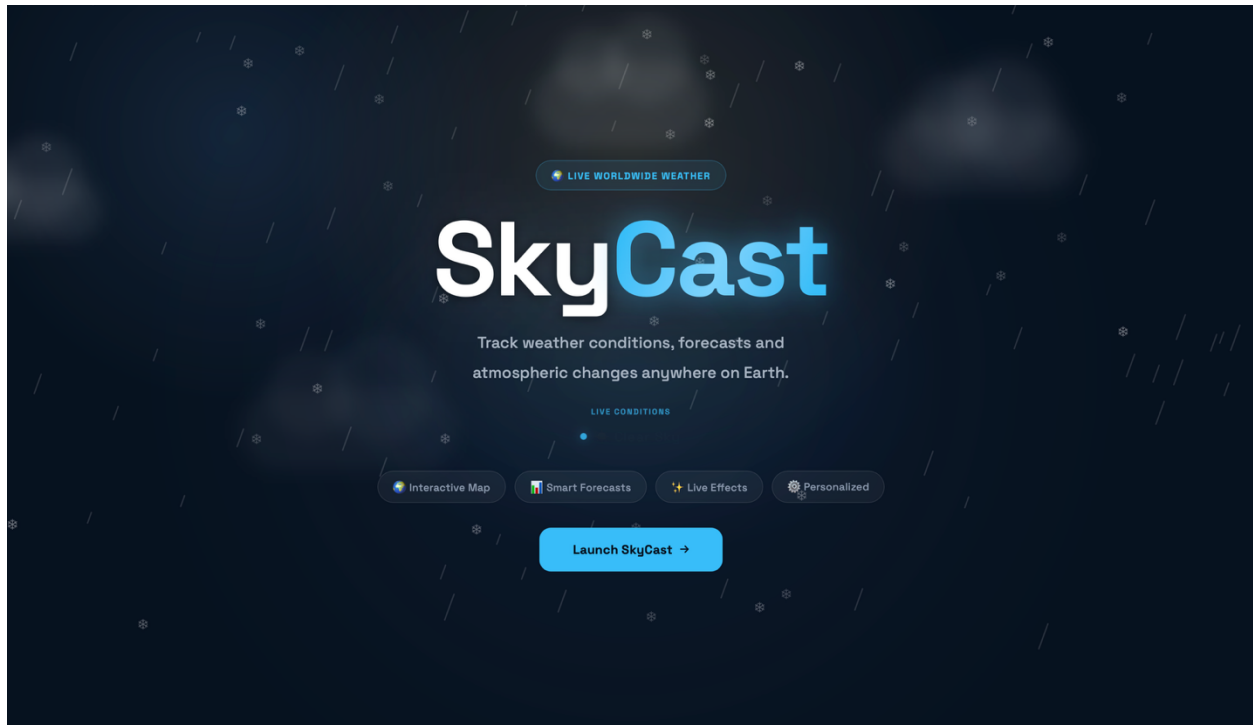
- Modern Landing Page
- Real-Time Weather Search
- Current Weather Dashboard
- Hourly Weather Forecast
- Five-Day Forecast
- Interactive Forecast Popups
- Dynamic Weather Icons
- Weather Insights
- Temperature Unit Toggle (°C / °F)
- Light & Dark Theme
- Interactive Weather Map
- City Search History
- Search Suggestions
- Today's Moment Recommendations

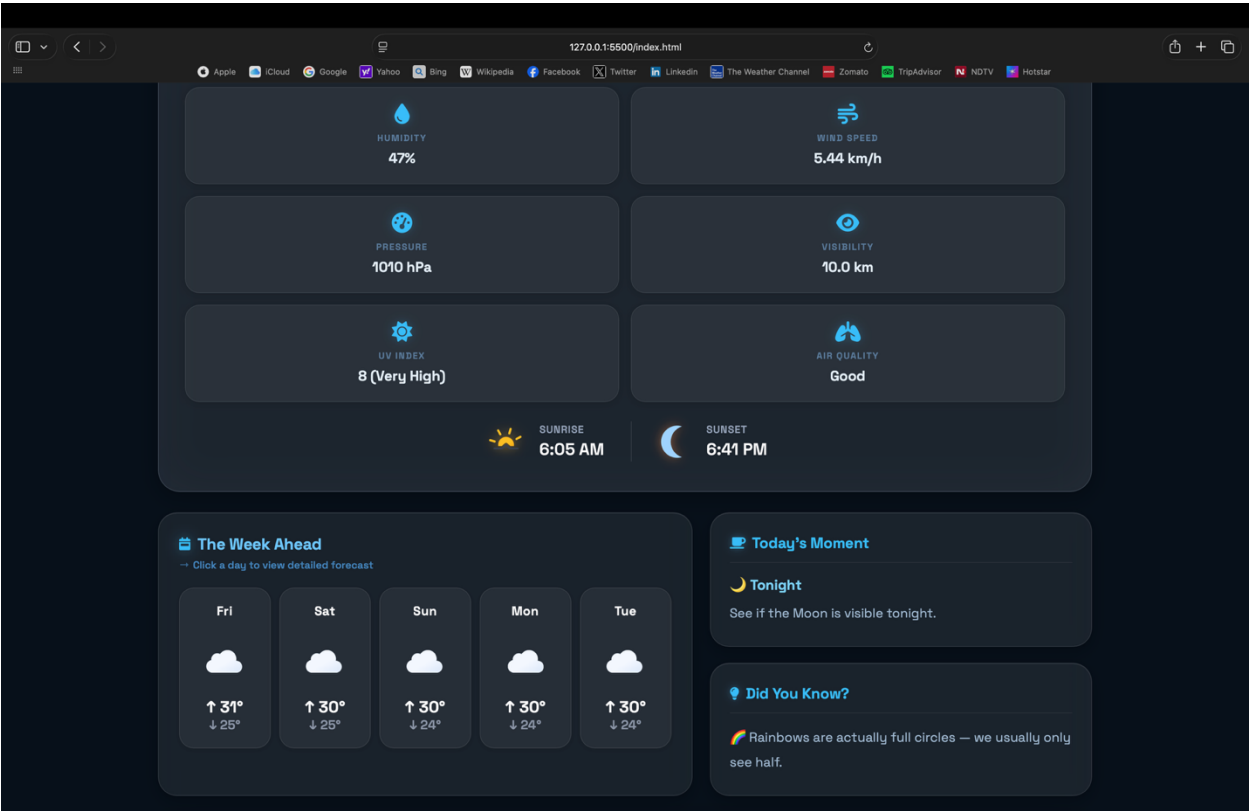
- Did You Know? Weather Facts
- Weather Effects Toggle
- Map Style Toggle
- Responsive Design
- User Settings Panel

Project Scope

The project serves as a complete weather forecasting platform that allows users to search weather information for cities worldwide. It provides real-time weather conditions, interactive forecasts, weather insights, and customization options within a modern user interface. The project demonstrates API integration, DOM manipulation, responsive web design, local storage management, and interactive frontend development techniques.

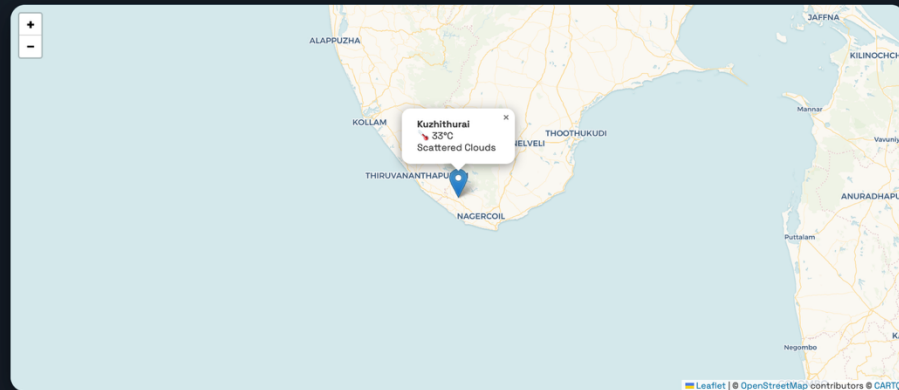
Screenshots





Global Weather Map

Click any location to explore its weather

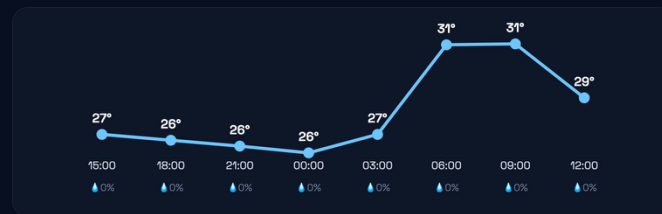


SkyCast

Weather made simple, beautiful and global.

© 2026 SKYCAST - POWERED BY OPENWEATHERMAP

Today & Tonight



Today's peak temperature: 9:00 AM (31°C)

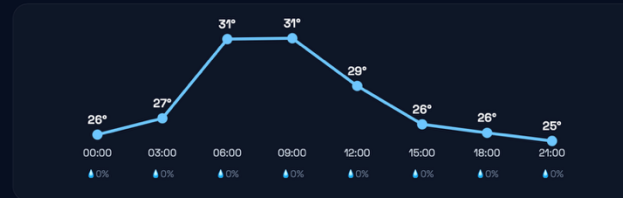
Sunset: 6:41 PM

Tonight's low: 26°C

Best time for a walk: 9:00 AM

No significant rain expected.

Friday Forecast



🔥 Warmest at 9:00 AM (31°C)

❄️ Coolest at 9:00 PM (25°C)

☁️ Highest rain chance at 9:00 PM (0%)

🌬️ Windiest at 9:00 AM (7 km/h)

🌞 Best time to go out: 6:00 PM

☀️ No significant rain expected.

Settings

APPEARANCE

🌙 Theme Dark ☐

PREFERENCES

🌡️ Temperature Units °C ☐

⚡ Weather Effects Off ☐

🗺️ Map Style Minimal ☐

🏠 Return To Landing Page



SkyCast

LIVE WORLDWIDE WEATHER



Find Weather Anywhere

Search over 200,000+ cities worldwide

Search

Cairo × California, Missouri, US × texas × Chennai, Tamil Nadu, IN ×

Gimbi

Thursday, June 25 at 9:51 AM

Updated just now

°C

°F



18°C

Light Rain

Feels Like 18°C

WET CONDITIONS



HUMIDITY
89%



WIND SPEED
0.80 km/h

The Week Ahead

→ Click a day to view detailed forecast

Fri



↑ 19°
↓ 13°

Sat



↑ 20°
↓ 12°

Sun



↑ 22°
↓ 12°

Mon



↑ 22°
↓ 12°

Tue



↑ 17°
↓ 13°

Today's Moment

Tiny Moment

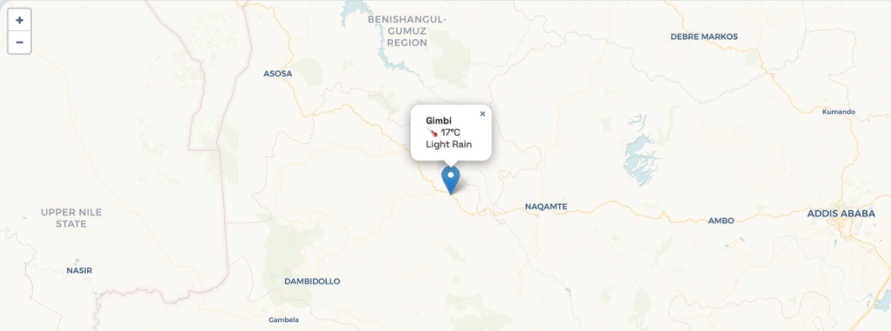
Notice where the sunlight falls today.

Did You Know?

Some clouds can stretch across several countries at once.

Global Weather Map

Click any location to explore its weather



 LIVE WORLDWIDE WEATHER

SkyCast

Track weather conditions, forecasts and atmospheric changes anywhere on Earth.

LIVE CONDITIONS

•  Clear Sky

 Interactive Map

 Smart Forecasts

✨ Live Effects

⚙️ Personalized

Launch SkyCast →



SkyCast



LIVE WORLDWIDE WEATHER

Find Weather Anywhere

Search over 200,000+ cities worldwide

 Search

Chennai 

 **Chennai**

Thursday, June 25 at 12:57 PM

 Updated just now

°C

°F



35°C

Overcast Clouds

Feels Like 37°C



COMFORTABLE CONDITIONS



HUMIDITY



WIND SPEED

Future Enhancements

- Severe weather alerts and notifications.
- Weather widgets for the home screen.
- Offline weather caching.
- Weather sharing through social media.
- Progressive Web App (PWA) support.

Conclusion

The SkyCast project successfully demonstrates the development of a modern, responsive weather forecasting web application using HTML, CSS, and JavaScript. By integrating real-time weather APIs, interactive maps, charts, customizable themes, and dynamic user interface components, the application delivers an engaging and informative user experience. Through this project, I strengthened my understanding of JavaScript, API integration, asynchronous programming, DOM manipulation, local storage, responsive web design, and frontend development best practices. SkyCast provides a clean, feature-rich, and user-friendly platform for accessing weather information worldwide.